Spontaneous Chirality and Nematicity Driven by Majorana Interactions in a Kitaev Spin Liquid

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The Kitaev spin liquid hosts fractional quasiparticles, including itinerant Majorana fermions and Z_2 fluxes. Symmetry-broken Kitaev spin liquids can realize distinct topological phases, including a Chern-number-zero phase with Abelian anyon topological order and a Chern-number-one phase with non-Abelian anyon topological order. These phases are conventionally understood to result from explicit symmetry breaking induced by external perturbations, such as strain or magnetic fields. Here, we show that analogous topological orders can emerge spontaneously from interactions among Majorana fermions. Using a mean-field analysis of an extended Kitaev model with four-spin interactions, we identify spontaneous chiral and nematic spin liquid phases through bulk Chern-number and edge-spectrum calculations. We also find a gapless Kitaev spin liquid regime with multiple Majorana cones. When a weak gap-opening perturbation is used as a diagnostic probe, this regime has a gapped descendant with higher Chern number. Our results show that Majorana interactions can reconstruct Majorana band topology in Kitaev spin liquids.

I. INTRODUCTION

The Kitaev honeycomb model is an exactly solvable quantum spin liquid in which spin degrees of freedom fractionalize into itinerant Majorana fermions and static Z_2 gauge fluxes [1–3]. In the flux-free ground-state sector [4], the model reduces to a free Majorana hopping problem on the honeycomb lattice. For the isotropic nearest-neighbor model, the Majorana spectrum is gapless and contains two inequivalent cones at the Brillouin-zone corners. When a gap-opening perturbation is present, the resulting quadratic Majorana band structure can be assigned an integer Chern number. In Kitaev's solution, this invariant determines the chirality of Majorana edge modes and is tied to the vortex statistics of the corresponding gapped Z_2 spin liquid [1, 5].

A central question is how this Majorana band structure can be modified beyond the nearest-neighbor Kitaev model. Previous studies have shown that the Majorana cones can be gapped or reorganized by magnetic fields [1, 6], by non-Kitaev exchange interactions in K-J-Γ-type models [7], and by perturbatively generated DM and Γ' interactions controlled by external fields or pressure [8]. Exactly solvable multispin extensions provide another route, where additional spin interactions generate prescribed further-neighbor Majorana hopping and realize higher-Chern-number sectors [5, 9]. In most of these settings, however, the additional mass term or hopping structure is introduced by an external perturbation or by a prescribed exactly solvable extension. A notable exception is the Kitaev model on the decorated (triangle-honeycomb) lattice, where the lattice geometry itself stabilizes a chiral spin liquid with spontaneous time-reversal symmetry breaking [10].

Here we study a different mechanism. We consider the Kitaev honeycomb model supplemented by a bond-pair four-spin interaction of strength J, measured in units of the Kitaev coupling K. The interaction couples two parallel Kitaev bond operators and commutes with the plaquette-flux operators W_p. It therefore leaves the Z_2 flux sectors conserved. Within the flux-free sector, the spin Hamiltonian reduces to the ordinary nearest-neighbor c-Majorana hopping term plus a quartic interaction among the matter Majoranas. Interacting Majorana lattice models provide a useful comparison because quartic terms can generate bilinear mean-field channels and interaction-driven topological band structures, as in the honeycomb Majorana-Hubbard model [11]. Our setting is different in an important respect. We do not add a bare further-neighbor Majorana hopping term to the flux-free Hamiltonian. Instead, the longer-range hopping and mass terms appear only as self-consistent mean-field channels of the four-spin interaction.

We analyze this model with a self-consistent mean-field theory restricted to the flux-free sector. The mean-field decoupling retains three bilinear channels of the quartic c-Majorana interaction. The P_α channel preserves the nearest-neighbor inter-sublattice form of the Kitaev term and changes its coefficient on an α bond from K to $K + 2JP_\alpha$. The Q_α channel produces a same-sublattice Haldane-type Majorana mass [12]. The S_a channel produces longer-range inter-sublattice hopping and changes the zero structure of the off-diagonal Majorana hopping amplitude. We use the anisotropy of P_α and S_a to diagnose nematicity, while the same-sublattice Q_α pattern diagnoses spontaneous time-reversal breaking.

Within this flux-free mean-field treatment, the phase diagram contains the ordinary gapless regime and three interaction-driven regimes. At large negative J/K, anisotropic P and S channels select a nematic Kitaev spin liquid with broken C_3 symmetry and $\mathcal{C} = 0$. At intermediate negative J/K, a spontaneous Q channel produces a chiral Kitaev spin liquid with broken time-reversal symmetry and $|\mathcal{C}| = 1$. These symmetry-broken regimes arise from the quartic Majorana interaction at zero field.

For positive J/K, the $\kappa = 0$ solution preserves C_3 and time-reversal symmetries while the P and S channels reconstruct the Majorana nodes. A quadratic touch-

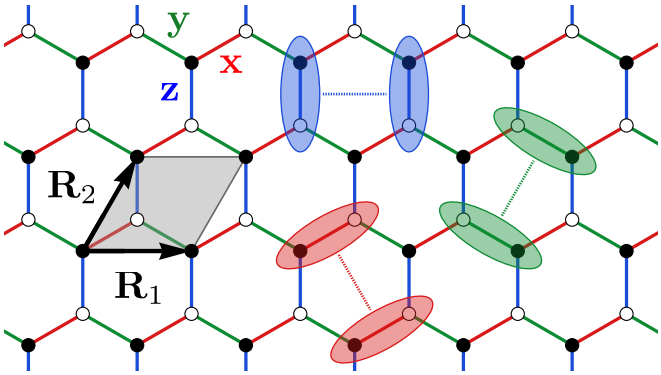


FIG. 1. Honeycomb-lattice convention and four-spin interaction. The unit cell contains one A site and one B site, with the x bond chosen as the intra-cell bond. The four-spin interaction J couples two parallel Kitaev bond operators separated by $\mathbf{p}_\alpha$.

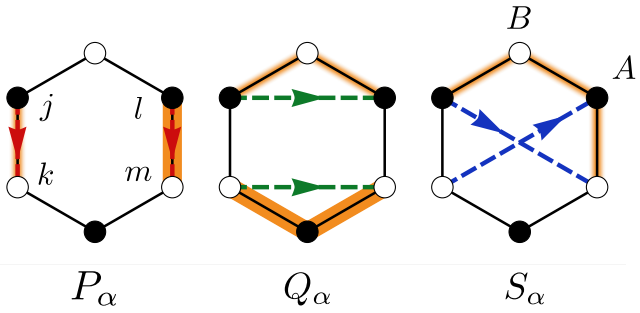


FIG. 2. Mean-field decoupling channels generated by the quartic interaction. The channel P_α changes the effective nearest-neighbor coefficient, Q_α gives an oriented same-sublattice mass, and S_a gives longer-range inter-sublattice hopping.

ing forms at Γ and splits into six additional Γ-centered Majorana cones, giving eight cones in total in the first Brillouin zone. Because the parent spectrum is gapless, Chern numbers are quoted only for weakly gapped descendants obtained by adding a small chiral mass after convergence. Across the reconstruction, the descendant changes from $\mathcal{C}_\kappa = -1$ to $\mathcal{C}_\kappa = +2$.

II. RESULTS

A. Model

We consider spin-1/2 moments on a honeycomb lattice with sublattices A and B. The bond index $\alpha = x, y, z$ labels the three Kitaev-bond types. We choose the x bond as the intra-cell bond and label each unit cell by the Bravais vector $\mathbf{r}$ of its A site. The spin Hamiltonian is

$$\begin{aligned} H &= H_K + H_J \\ &= K \sum_{\mathbf{r},\alpha} \sigma^\alpha_{A,\mathbf{r}} \sigma^\alpha_{B,\mathbf{r}+\boldsymbol{\Delta}_\alpha} \\ &\quad + J \sum_{\mathbf{r},\alpha} \left(\sigma^\alpha_{A,\mathbf{r}} \sigma^\alpha_{B,\mathbf{r}+\boldsymbol{\Delta}_\alpha}\right) \left(\sigma^\alpha_{A,\mathbf{r}+\mathbf{p}_\alpha} \sigma^\alpha_{B,\mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha}\right). \end{aligned} \tag{1}$$

Here $\sum_{\mathbf{r},\alpha}$ denotes a sum over unit cells and over the three bond types. In a single term of H_J, we denote the four sites by $j = (A, \mathbf{r})$, $k = (B, \mathbf{r}+\boldsymbol{\Delta}_\alpha)$, $\ell = (A, \mathbf{r}+\mathbf{p}_\alpha)$, and $m = (B, \mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha)$. Thus j, k and ℓ, m form two parallel α bonds. We use $\mathbf{R}_1 = (\sqrt{3}, 0)$ and $\mathbf{R}_2 = (\sqrt{3}/2, 3/2)$. The A-to-B cell displacements are $\boldsymbol{\Delta}_x = \mathbf{0}$, $\boldsymbol{\Delta}_y = -\mathbf{R}_1$, and $\boldsymbol{\Delta}_z = -\mathbf{R}_2$. The bond-pair separations are $\mathbf{p}_x = \mathbf{R}_1 - \mathbf{R}_2$, $\mathbf{p}_y = \mathbf{R}_2$, and $\mathbf{p}_z = \mathbf{R}_1$. The geometry is shown in Fig. 1.

We use the Kitaev Majorana representation $\sigma^\alpha_i = i b^\alpha_i c_i$. For an A-to-B α bond with $j = (A, \mathbf{r})$ and $k = (B, \mathbf{r}+\boldsymbol{\Delta}_\alpha)$, we define $u^\alpha_{BA}(\mathbf{r}) = i b^\alpha_k b^\alpha_j$ and $u^\alpha_{AB}(\mathbf{r}) = -u^\alpha_{BA}(\mathbf{r})$. We denote $u^\alpha_{BA}(\mathbf{r})$ by $u^\alpha_{\mathbf{r}}$. Then $\sigma^\alpha_{A,\mathbf{r}} \sigma^\alpha_{B,\mathbf{r}+\boldsymbol{\Delta}_\alpha} = i u^\alpha_{\mathbf{r}} c_{A,\mathbf{r}} c_{B,\mathbf{r}+\boldsymbol{\Delta}_\alpha}$. For a hexagon p, the plaquette operator is $W_p = \sigma^x_1 \sigma^y_2 \sigma^z_3 \sigma^x_4 \sigma^y_5 \sigma^z_6 = \pm 1$, equivalently the oriented product of the six static gauge variables around p. The bond-pair term in Eq. (1) preserves the W_p eigenvalues.

We restrict the calculation to the translationally invariant flux-free sector $W_p = +1$ and choose the uniform gauge $u^\alpha_{BA}(\mathbf{r}) = u^\alpha_{\mathbf{r}} = 1$, or equivalently $u^\alpha_{AB}(\mathbf{r}) = -1$. In this sector, the matter-Majorana Hamiltonian is

$$\begin{aligned} H &= H_K + H_J \\ &= K \sum_{\mathbf{r},\alpha} i c_{A,\mathbf{r}} c_{B,\mathbf{r}+\boldsymbol{\Delta}_\alpha} \\ &\quad + J \sum_{\mathbf{r},\alpha} \left[i c_{A,\mathbf{r}} c_{B,\mathbf{r}+\boldsymbol{\Delta}_\alpha}\right] \left[i c_{A,\mathbf{r}+\mathbf{p}_\alpha} c_{B,\mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha}\right]. \end{aligned} \tag{2}$$

The first term is the nearest-neighbor matter-Majorana hopping of the Kitaev model, while the second term is quartic in the matter Majoranas. Thus the flux-free problem is interacting and is treated below using self-consistent Majorana mean-field theory. Possible competition with other flux sectors is left outside the present flux-free mean-field treatment.

B. Mean-field channels

The quartic term in Eq. (2) generates three inequivalent bilinear channels. We keep the direct inter-sublattice channel P_α, the oriented same-sublattice channel Q_α, and the crossed inter-sublattice channel S_a, defined by

$$\begin{aligned} P_\alpha &= \langle i c_{A,\mathbf{r}} c_{B,\mathbf{r}+\boldsymbol{\Delta}_\alpha} \rangle, \\ Q_\alpha &= \langle i c_{A,\mathbf{r}} c_{A,\mathbf{r}+\mathbf{p}_\alpha} \rangle = -\langle i c_{B,\mathbf{r}} c_{B,\mathbf{r}+\mathbf{p}_\alpha} \rangle, \\ S_a &= \langle i c_{A,\mathbf{r}} c_{B,\mathbf{r}+\boldsymbol{\delta}_a} \rangle. \end{aligned} \tag{3}$$

Here $\alpha = x, y, z$, while $a = 1, 2, 3$ labels the longer inter-sublattice displacements $\boldsymbol{\delta}_1 = \mathbf{R}_1 - \mathbf{R}_2$, $\boldsymbol{\delta}_2 = \mathbf{R}_2 - \mathbf{R}_1$, and $\boldsymbol{\delta}_3 = -\mathbf{R}_1 - \mathbf{R}_2$. The minus sign in the definition of Q_α fixes the relative orientation of same-sublattice hopping on the two sublattices. A C_3-symmetric chiral channel is described by a single amplitude Q_χ with $(Q_z, Q_x, Q_y) = Q_\chi(+1, -1, -1)$, and time reversal changes $Q_\chi \to -Q_\chi$.

The three channels enter the quadratic problem in different ways. The P_α channel changes the nearest-neighbor coefficients from K to $K + 2JP_\alpha$. The Q_α channel gives a diagonal Haldane-type mass. The S_a channel gives longer-range inter-sublattice hopping and changes the zeros of the off-diagonal hopping amplitude $\Phi(\mathbf{k})$. These channels are illustrated in Fig. 2.

To fix the phases in the momentum-space Hamiltonian, we use

$$c_{\lambda,\mathbf{k}} = N_{\mathrm{uc}}^{-1/2} \sum_{\mathbf{r}} e^{-i\mathbf{k}\cdot\mathbf{r}} c_{\lambda,\mathbf{r}}, \qquad \lambda = A, B,$$
$$\Psi_{\mathbf{k}} = (c_{A,\mathbf{k}}, c_{B,\mathbf{k}})^T.$$

The mean-field Hamiltonian is

$$H_{\mathrm{MF}} = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger \mathcal{H}(\mathbf{k}) \Psi_{\mathbf{k}} + E_{\mathrm{const}},$$
$$\mathcal{H}(\mathbf{k}) = \begin{pmatrix} -m(\mathbf{k}) & i\Phi(\mathbf{k}) \\ -i\Phi^*(\mathbf{k}) & m(\mathbf{k}) \end{pmatrix}, \tag{4}$$
$$m(\mathbf{k}) = m_J(\mathbf{k}) + m_\kappa(\mathbf{k}).$$

The off-diagonal hopping amplitude is

$$\begin{aligned}\Phi(\mathbf{k}) =& (K + 2JP_x) + (K + 2JP_y) e^{-i\mathbf{k}\cdot\mathbf{R}_1} \\ &+ (K + 2JP_z) e^{-i\mathbf{k}\cdot\mathbf{R}_2} - J(S_2 + S_3) e^{+i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)} \\ &- J(S_1 + S_3) e^{-i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)} \\ &- J(S_1 + S_2) e^{-i\mathbf{k}\cdot(\mathbf{R}_1 + \mathbf{R}_2)}. \end{aligned} \tag{5}$$

The two terms in the diagonal mass are

$$\begin{aligned} m_J(\mathbf{k}) =& 2J\big[Q_z \sin(\mathbf{k}\cdot\mathbf{R}_1) + Q_x \sin\{\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\} \\ &+ Q_y \sin(\mathbf{k}\cdot\mathbf{R}_2)\big], \end{aligned} \tag{6}$$
$$\begin{aligned} m_\kappa(\mathbf{k}) =& 2\kappa\big[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\{\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\} \\ &- \sin(\mathbf{k}\cdot\mathbf{R}_2)\big]. \end{aligned} \tag{7}$$

The self-consistency problem is solved with $m_\kappa = 0$, so that

$$\omega_0(\mathbf{k}) = \sqrt{m_J(\mathbf{k})^2 + |\Phi(\mathbf{k})|^2}. \tag{8}$$

The nine quantities

$$X = (P_z, P_x, P_y, Q_z, Q_x, Q_y, S_1, S_2, S_3)$$

are determined self-consistently at $\kappa = 0$. The explicit fixed-point equations and the energy selection among converged solutions are given in Sec. IV. The mass $m_\kappa(\mathbf{k})$ is added only after the $\kappa = 0$ solution is fixed, when the gapped Chern diagnostic is evaluated. Chern labels assigned to gapless regimes therefore refer to weakly gapped descendants. The phase-diagram labels and bulk Chern diagnostics use $\kappa = 0.02$, while the ribbon spectra use the finite κ stated in the corresponding caption.

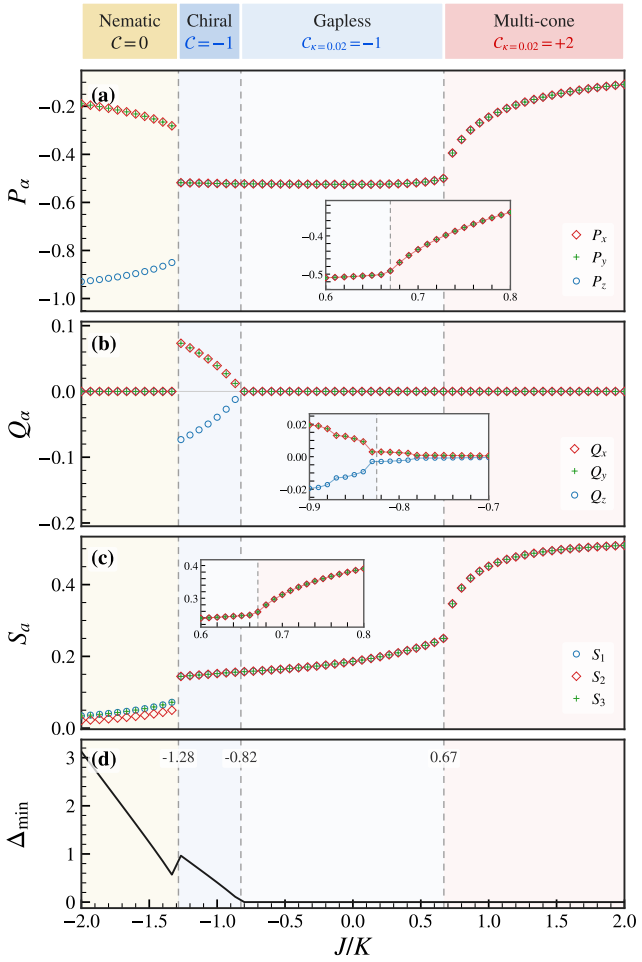


FIG. 3. Self-consistent phase diagram. The mean-field channels P_α, Q_α, and S_a, and the minimum Majorana gap $\Delta_{\min}$, are shown as functions of J/K at $\kappa = 0$. The four regimes are the nematic, chiral, ordinary gapless, and multi-cone Kitaev spin liquids. The labels $\mathcal{C} = 0$ and $\mathcal{C} = -1$ refer to gapped $\kappa = 0$ Hamiltonians. The labels $\mathcal{C}_\kappa = -1$ and $\mathcal{C}_\kappa = +2$ refer to weakly gapped descendants obtained by adding the diagnostic mass m_κ after the $\kappa = 0$ self-consistent solution is fixed.

C. Phase diagram

We solve Eq. (16) as a function of J/K, setting $K = 1$ throughout the numerical results. Figure 3 shows the resulting P_α, Q_α, and S_a channels, together with the minimum Majorana gap $\Delta_{\min} = 2 \min_{\mathbf{k}} \omega_0(\mathbf{k})$. The calculation is performed at $\kappa = 0$, so the plotted channels are selected by the bond-pair interaction itself. The scan

gives four regimes separated near $J/K \simeq -1.28$, -0.82, and 0.67. The first two regimes are gapped already at $\kappa = 0$. The latter two remain gapless at $\kappa = 0$, and are distinguished by the structure of their Majorana nodes.

Nematic KSL. For $J/K \lesssim -1.28$, the self-consistent solution is anisotropic in the off-diagonal channels. In the representative solution shown in Fig. 3, $P_z \neq P_x = P_y$, $S_1 = S_3 \neq S_2$, and $Q_\alpha \simeq 0$. This pattern selects one Kitaev-bond direction and spontaneously breaks lattice C_3 rotation symmetry. Since the Q channel is absent, the gap is generated by the anisotropic off-diagonal hopping amplitude $\Phi(\mathbf{k})$. The resulting Majorana band has $\mathcal{C} = 0$.

Chiral KSL. At $J/K \simeq -1.28$, the solution changes discontinuously. For $-1.28 \lesssim J/K \lesssim -0.82$, the P and S channels are C_3-symmetric, while the oriented same-sublattice channel develops $Q_\chi \neq 0$. Equivalently, $P_x = P_y = P_z$, $S_1 = S_2 = S_3$, and $Q_z = -Q_x = -Q_y$. A nonzero Q_χ gives a Haldane-type Majorana mass and opens a gap without an external magnetic field. The plotted time-reversal domain has $\mathcal{C} = -1$. The opposite domain has $Q_\chi \to -Q_\chi$ and $\mathcal{C} = +1$.

Gapless KSL. For $-0.82 \lesssim J/K \lesssim 0.67$, the spontaneous Q channel vanishes within numerical accuracy. The solution preserves C_3, and the Majorana spectrum is gapless at $\kappa = 0$. Therefore the Chern number is not defined for the $\kappa = 0$ Hamiltonian. When a weak diagnostic mass m_κ is added after the self-consistent solution is obtained, the corresponding weakly gapped descendant has $\mathcal{C}_\kappa = -1$.

Multi-cone KSL. For $J/K \gtrsim 0.67$, the solution remains C_3-symmetric and the spontaneous Q channel vanishes within numerical accuracy. The Majorana spectrum is gapless at $\kappa = 0$, but its low-energy nodes are reconstructed by the appearance of six additional Γ-centered Majorana cones. Together with the original K and K' cones, the multi-cone spectrum contains eight cones in the first Brillouin zone. Therefore the Chern number is not defined for the $\kappa = 0$ Hamiltonian. The weakly gapped descendant obtained by adding m_κ, under the same diagnostic mass convention, has $\mathcal{C}_\kappa = +2$.

Figure 4 visualizes the same sequence at the band-structure level. The first negative-J transition is first order, while the positive-J transition is a nodal reconstruction through a Γ-point quadratic touching.

D. Symmetry Diagnostics

After the self-consistency equations are solved, the mean-field amplitudes provide a compact symmetry diagnostic of the flux-free solution. Anisotropy among P_α or S_a detects spontaneous C_3 breaking, while the oriented same-sublattice component Q_χ detects spontaneous time-reversal breaking. When these diagnostics vanish, the remaining distinction between the two positive-J gapless regimes is nodal rather than symmetry-breaking.

TABLE I. Diagnostics used to label the self-consistent regimes. The column "SSB" marks spontaneous symmetry breaking detected by the mean-field amplitudes. The multi-cone row denotes six additional cones surrounding Γ, not cones pinned at Γ.

Regime	SSB	Nodes/gap	Criterion
Nematic	C_3	gapped	$\mathcal{N}_P$ or $\mathcal{N}_S \neq 0$
Chiral	$\mathcal{T}$	gapped	$Q_\chi \neq 0$
Gapless	none	K, K'	$\delta_\Gamma > 0$
Multi-cone	none	$K, K' + 6$ near Γ	$\delta_\Gamma < 0$

The C_3-breaking diagnostics are

$$\mathcal{N}_P = \left\{ \frac{1}{3} \left[(P_z - P_x)^2 + (P_x - P_y)^2 + (P_y - P_z)^2 \right] \right\}^{1/2}, \tag{9}$$

$$\mathcal{N}_S = \left\{ \frac{1}{3} \left[(S_1 - S_2)^2 + (S_2 - S_3)^2 + (S_3 - S_1)^2 \right] \right\}^{1/2}. \tag{10}$$

They vanish for a C_3-symmetric solution and become finite when one lattice direction is selected spontaneously.

The time-reversal-odd same-sublattice component is

$$Q_\chi = \frac{1}{3} (Q_z - Q_x - Q_y). \tag{11}$$

For a C_3-symmetric chiral solution, $(Q_z, Q_x, Q_y) = Q_\chi(+1, -1, -1)$. Thus $Q_\chi \neq 0$ diagnoses spontaneous $\mathcal{T}$ breaking.

When $\mathcal{N}_P = \mathcal{N}_S = 0$ and $Q_\chi \simeq 0$, the C_3-symmetric sector is further labeled by

$$\delta_\Gamma \equiv \frac{\Phi(\Gamma)}{3} = K + 2J(P - S), \tag{12}$$

where $P_x = P_y = P_z = P$ and $S_1 = S_2 = S_3 = S$. The sign of δ_Γ separates the ordinary gapless and multi-cone regimes. Its band-theory content is analyzed in Sec. II E.

E. Band Topology

The two positive-J gapless regimes have the same C_3 and time-reversal diagnostics, but differ in their matter-Majorana nodal content. This type of distinction is familiar in proximate Kitaev spin liquids, where states with the same symmetry data can differ by their Majorana cone content and by the Chern response obtained after a weak time-reversal-breaking perturbation is applied [7, 8]. We therefore first analyze the $\kappa = 0$ nodal structure and then add a small m_κ term, with P, Q, S fixed at their $\kappa = 0$ self-consistent values, to define the associated gapped Chern diagnostic.

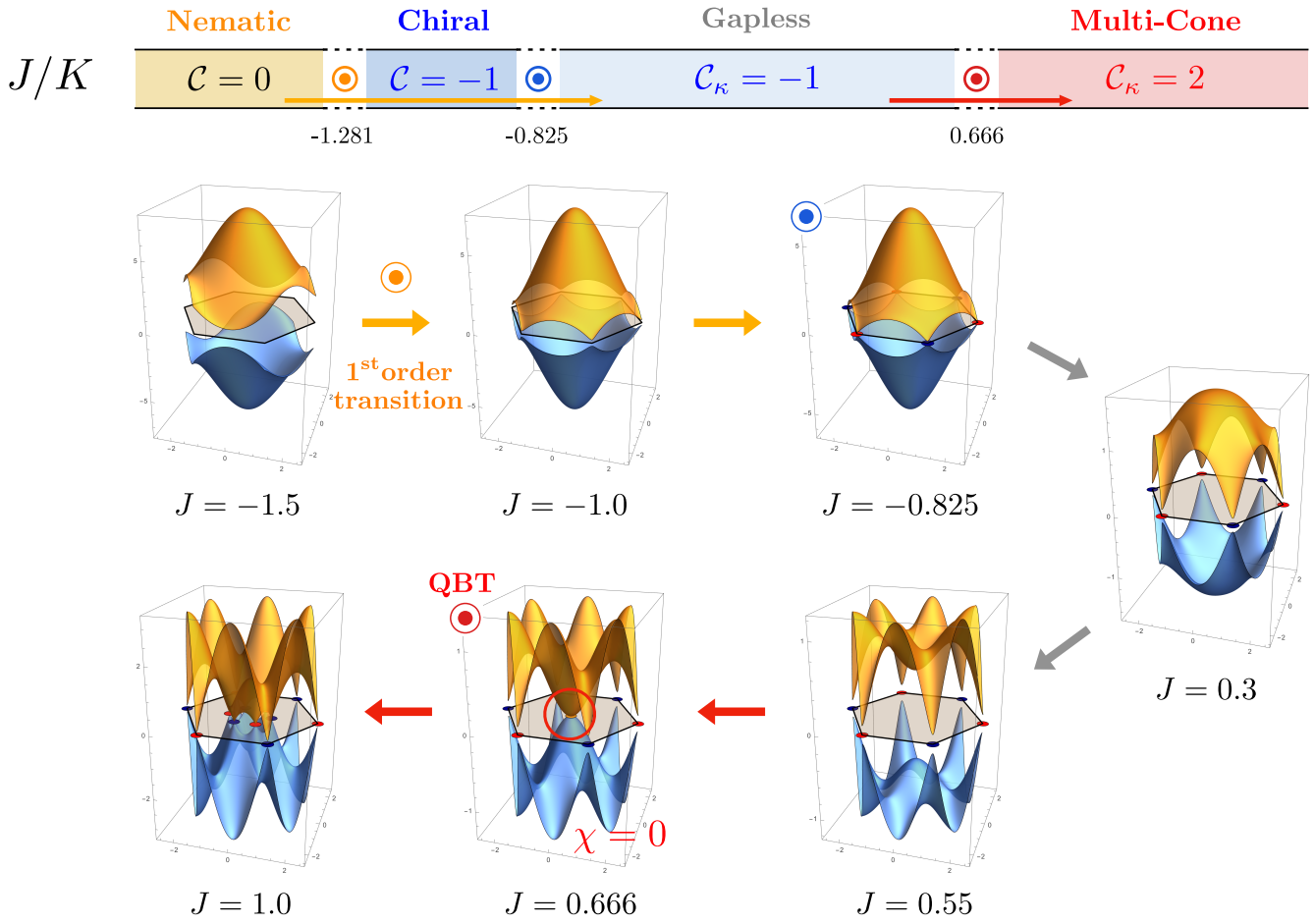


FIG. 4. Evolution of the self-consistent Majorana bands across the four regimes. The upper strip summarizes the phase sequence and the Chern labels used in Fig. 3. The lower panels show representative bulk dispersions: the nematic and chiral phases are gapped, the ordinary positive-J regime is gapless, and the multi-cone regime is reached through a Γ-point quadratic band touching near $J/K \simeq 0.67$.

In the C_3-symmetric sector with $Q_\chi = 0$ and $m_\kappa = 0$,

$$\mathcal{H}_0(\mathbf{k}) = \begin{pmatrix} 0 & i\Phi(\mathbf{k}) \\ -i\Phi^*(\mathbf{k}) & 0 \end{pmatrix}, \qquad E_\pm(\mathbf{k}) = \pm|\Phi(\mathbf{k})|.$$

The nodes are zeros of $\Phi(\mathbf{k})$. The self-consistent P and S channels therefore reconstruct the nodal set through the off-diagonal kernel, rather than by generating a mass.

The reconstruction is controlled by

$$\Phi(\Gamma) = 3\delta_\Gamma, \qquad \delta_\Gamma = K + 2J(P - S).$$

At $\delta_\Gamma = 0$, the constant and linear terms of the Γ-point expansion vanish, producing a quadratic band touching at Γ. On the multi-cone side, the same expansion resolves the touching into six simple zeros surrounding Γ. Together with the original K, K' nodes, this gives the multi-cone spectrum summarized in Table I. The local expansion and winding assignment are collected in Appendix C.

For the gapped Hamiltonian obtained by adding m_κ, we evaluate the lower-band Chern number as

$$\mathcal{C}_\kappa = \frac{1}{2\pi}\int_{\mathrm{BZ}} d^2k\, \Omega_-(\mathbf{k}),$$
$$\Omega_-(\mathbf{k}) = -2\,\mathrm{Im}\,\frac{M_x(\mathbf{k})M_y(\mathbf{k})}{(E_+ - E_-)^2}, \tag{13}$$

where

$$M_x(\mathbf{k}) = \langle u_-|\partial_{k_x}\mathcal{H}|u_+\rangle,$$
$$M_y(\mathbf{k}) = \langle u_+|\partial_{k_y}\mathcal{H}|u_-\rangle. \tag{14}$$

Here $|u_\pm\rangle$ and $E_\pm$ are the two eigenstates and eigenvalues of the m_κ-gapped two-band Hamiltonian. The full Brillouin-zone integral fixes the values quoted in Fig. 3, while the cone decomposition resolves the jump across the reconstruction.

The jump in $\mathcal{C}_\kappa$ follows from the massive cones created by the split Γ-point touching. A simple zero $\mathbf{k}_n$ of Φ,

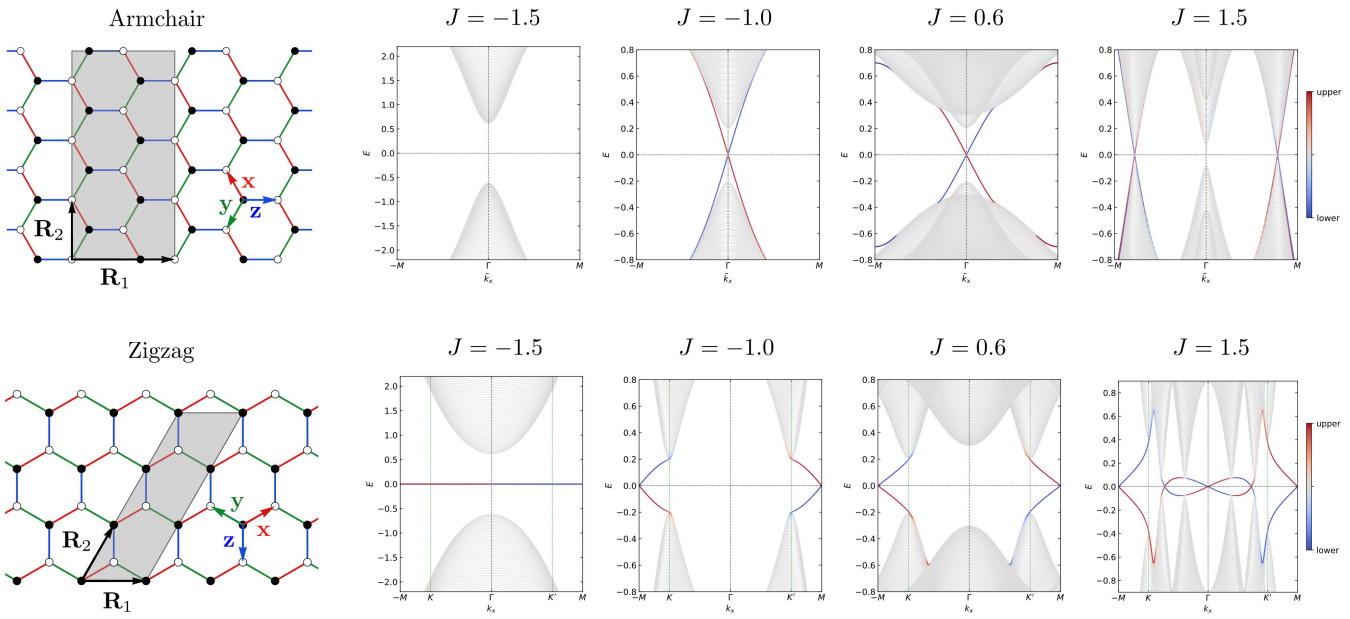


FIG. 5. Representative ribbon spectra for armchair and zigzag edges. The four columns correspond to the nematic, chiral, ordinary gapless-descendant, and multi-cone-descendant regimes. Grey states form the bulk projection, while colored branches mark edge-localized modes on opposite boundaries. The spectra are computed from finite-width real-space Hamiltonians using the converged bulk mean-field parameters; in the gapless regimes a small diagnostic mass is included to display the gapped descendant.

with winding w_n and mass $m_n = m_\kappa(\mathbf{k}_n)$, contributes

$$C_n = -\frac{1}{2} w_n \operatorname{sgn} m_n.$$

For the six cones generated from the Γ-point touching, the expansion in Appendix C gives a total

$$\Delta\mathcal{C}_\kappa = 3 \qquad (\kappa > 0).$$

Combining this local result with the full Brillouin-zone calculation gives $\mathcal{C}_\kappa = -1$ for the ordinary gapless regime after m_κ is added, and $\mathcal{C}_\kappa = +2$ for the corresponding multi-cone regime. The self-consistent $\kappa = 0$ solutions themselves remain gapless.

The corresponding edge spectra are shown in Fig. 5. The gapped phases display the expected absence or presence of chiral Majorana edge modes, while the weakly gapped descendants of the two positive-J regimes show the change in edge-mode count associated with $\mathcal{C}_\kappa = -1$ and $\mathcal{C}_\kappa = +2$.

F. Microscopic origin

III. DISCUSSION

We have studied a four-spin extension of the Kitaev honeycomb model that preserves the plaquette flux operators. In the uniform flux-free gauge, this interaction becomes a quartic interaction of the itinerant c Majorana fermions. Its mean-field decoupling yields three bilinear amplitudes. The P_α channel preserves the nearest-neighbor inter-sublattice Kitaev form and changes the corresponding bond coefficient from K to $K + 2JP_\alpha$. The Q_α channel gives a same-sublattice Haldane mass, and the S_a channel gives longer-range inter-sublattice hopping. The Q_α and S_a terms are not independent quadratic perturbations, but arise from the self-consistent decoupling of the quartic Majorana interaction.

The negative-J phases are selected by symmetry-breaking channels. In the nematic phase, P_α and S_a become anisotropic and choose one lattice direction. In the chiral phase, the C_3-symmetric solution develops $Q_\chi \neq 0$, which gaps the K, K' cones with a Haldane-type mass. This chiral mass is spontaneous: it appears in a time-reversal-symmetric spin Hamiltonian, unlike the explicit field-induced mass of the pure Kitaev model. This mechanism is related to interaction-driven topology in Majorana-Hubbard models, but here the time-reversal breaking appears as a self-consistent Q-channel instability [10, 11].

The positive-J transition has a different character. The ordinary gapless and multi-cone regimes have the same C_3 and time-reversal diagnostics within the flux-free mean-field solution. Their difference lies in the nodal structure of the off-diagonal Majorana kernel $\Phi(\mathbf{k})$. The self-consistent P and S channels tune

$$\delta_\Gamma = K + 2J(P - S).$$

At $\delta_\Gamma = 0$, the Majorana spectrum has a quadratic band touching at Γ. On the multi-cone side, this touching splits into six simple cones surrounding Γ, while the original K, K' cones remain. Thus the multi-cone regime is not identified by a new local order parameter. It is identified by a symmetry-preserving reconstruction of the matter-Majorana nodes, as in gapless Kitaev-like spin liquids whose symmetry data do not fix the number of Majorana cones [7, 8].

After the $\kappa = 0$ self-consistent parameters are fixed, adding a small time-reversal-breaking mass $m_\kappa(\mathbf{k})$ gives $\mathcal{C}_\kappa = -1$ for the ordinary gapless regime and $\mathcal{C}_\kappa = +2$ for the multi-cone regime. The difference comes from the six cones produced by the split Γ-point touching. Their windings and m_κ mass signs give

$$\Delta\mathcal{C}_\kappa = 3.$$

The $\kappa = 0$ solutions remain gapless; the Chern labels therefore refer to the Hamiltonian after the small m_κ term is added. The ribbon spectra show the corresponding change in the number of chiral Majorana edge modes for the gapped Hamiltonians being plotted.

Several extensions remain. The stability of these mean-field regimes should be tested beyond the uniform flux-free sector, and physical spin responses should be computed. If a time-reversal-breaking perturbation realizes m_κ, the change from $\mathcal{C}_\kappa = -1$ to $+2$ gives a target for edge and thermal-transport diagnostics. Thus the model provides a minimal mean-field setting for studying nematicity, spontaneous chirality, and symmetry-preserving Majorana nodal reconstruction generated by one flux-preserving interaction.

IV. METHODS

We solve the quartic matter-Majorana problem in the uniform flux-free sector. The self-consistent parameters are determined at $\kappa = 0$. The explicit mass m_κ is not included in the fixed-point equation and is added only when the gapped Chern diagnostic is evaluated.

For each four-Majorana term, with sites denoted by j, k, ℓ, m, the three inequivalent contractions are

$$(ic_jc_k)(ic_\ell c_m), \qquad -(ic_jc_\ell)(ic_kc_m), \qquad (ic_jc_m)(ic_kc_\ell).$$

They define the direct inter-sublattice channel P_α, the same-sublattice channel Q_α, and the crossed inter-sublattice channel S_a. With

$$X = (P_z, P_x, P_y, Q_z, Q_x, Q_y, S_1, S_2, S_3),$$

the mean-field Hamiltonian takes the two-band form given in Sec. II B. The real-space sign convention and the constant terms are fixed in Appendix A.

For a fixed X, the mean-field energy density is

$$\begin{aligned}\varepsilon_{\mathrm{MF}}(X) = &- J \sum_{\alpha=z,x,y} P_\alpha^2 - J \sum_{\alpha=z,x,y} Q_\alpha^2 \\ &+ J(S_1S_2 + S_1S_3 + S_2S_3) - \frac{1}{N_k}\sum_{\mathbf{k}} \omega_0(\mathbf{k}).\end{aligned} \tag{15}$$

The fixed-point equations are obtained by evaluating the same bilinears in the occupied lower band. Equivalently, they are the stationarity conditions of $\varepsilon_{\mathrm{MF}}(X)$ with respect to the nine bilinear parameters. The explicit map is

$$\begin{aligned} P_\alpha^{\mathrm{out}} &= -\frac{1}{N_k}\sum_{\mathbf{k}} \frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot\boldsymbol{\Delta}_\alpha}\Phi^*(\mathbf{k})\right]}{\omega_0(\mathbf{k})}, \\ S_a^{\mathrm{out}} &= -\frac{1}{N_k}\sum_{\mathbf{k}} \frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot\boldsymbol{\delta}_a}\Phi^*(\mathbf{k})\right]}{\omega_0(\mathbf{k})}, \\ Q_\alpha^{\mathrm{out}} &= -\frac{1}{N_k}\sum_{\mathbf{k}} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha)\frac{m_J(\mathbf{k})}{\omega_0(\mathbf{k})}. \end{aligned} \tag{16}$$

A self-consistent solution satisfies $X = X^{\mathrm{out}}(X)$.

The fixed-point equation is solved by linear mixing. For each value of J/K, we use several initial conditions, including nearby converged solutions and symmetry-biased seeds for the nematic and chiral regimes. When more than one stationary solution is obtained, the phase diagram uses the solution with the lowest $\varepsilon_{\mathrm{MF}}$.

After the $\kappa = 0$ fixed point is obtained, $m_\kappa(\mathbf{k})$ is added to compute the lower-band Chern number of the corresponding gapped two-band Hamiltonian. The full Brillouin-zone Chern number is the primary topological diagnostic, while cone-resolved windings are used only to interpret changes in the integrated value. Ribbon spectra are computed from finite-width real-space Hamiltonians built with the converged bulk parameters.

Appendix A: Mean-field kernel

In the uniform flux-free sector, the four-spin interaction becomes

$$H_J = J \sum_{\alpha=x,y,z} \sum_{\mathbf{r}} \left[ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\Delta}_\alpha)\right]\left[ic_A(\mathbf{r}+\mathbf{p}_\alpha)c_B(\mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha)\right], \tag{A1}$$

where $\boldsymbol{\Delta}_x = \mathbf{0}$, $\boldsymbol{\Delta}_y = -\mathbf{R}_1$, $\boldsymbol{\Delta}_z = -\mathbf{R}_2$, and $\mathbf{p}_x = \mathbf{R}_1 - \mathbf{R}_2$, $\mathbf{p}_y = \mathbf{R}_2$, $\mathbf{p}_z = \mathbf{R}_1$. Explicitly,

$$\begin{aligned}
H_J^{(x)} &= J\sum_{\mathbf{r}} \big[ic_A(\mathbf{r})c_B(\mathbf{r})\big]\big[ic_A(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)c_B(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)\big],\\
H_J^{(y)} &= J\sum_{\mathbf{r}} \big[ic_A(\mathbf{r})c_B(\mathbf{r}-\mathbf{R}_1)\big]\big[ic_A(\mathbf{r}+\mathbf{R}_2)c_B(\mathbf{r}+\mathbf{R}_2-\mathbf{R}_1)\big],\\
H_J^{(z)} &= J\sum_{\mathbf{r}} \big[ic_A(\mathbf{r})c_B(\mathbf{r}-\mathbf{R}_2)\big]\big[ic_A(\mathbf{r}+\mathbf{R}_1)c_B(\mathbf{r}+\mathbf{R}_1-\mathbf{R}_2)\big].
\end{aligned} \tag{A2}$$

For a fixed α, denote the four Majoranas in Eq. (A1) by

$$\begin{aligned}
j &= A(\mathbf{r}), & k &= B(\mathbf{r}+\boldsymbol{\Delta}_\alpha),\\
\ell &= A(\mathbf{r}+\mathbf{p}_\alpha), & m &= B(\mathbf{r}+\mathbf{p}_\alpha+\boldsymbol{\Delta}_\alpha).
\end{aligned} \tag{A3}$$

The three inequivalent pairings of $(ic_jc_k)(ic_\ell c_m)$ are

$$(jk)(\ell m), \qquad -(j\ell)(km), \qquad (jm)(k\ell), \tag{A4}$$

where the relative signs follow from Majorana anticommutation. These pairings define

$$\begin{aligned}
P_\alpha &= \langle ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\Delta}_\alpha)\rangle\,,\\
Q_\alpha &= \langle ic_A(\mathbf{r})c_A(\mathbf{r}+\mathbf{p}_\alpha)\rangle = -\langle ic_B(\mathbf{r})c_B(\mathbf{r}+\mathbf{p}_\alpha)\rangle\,,\\
S_a &= \langle ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\delta}_a)\rangle\,,
\end{aligned} \tag{A5}$$

with $\boldsymbol{\delta}_1 = \mathbf{R}_1 - \mathbf{R}_2$, $\boldsymbol{\delta}_2 = \mathbf{R}_2 - \mathbf{R}_1$, and $\boldsymbol{\delta}_3 = -\mathbf{R}_1 - \mathbf{R}_2$.

Using

$$c_X(\mathbf{r}) = \frac{1}{\sqrt{N}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} c_X(\mathbf{k}), \qquad X = A, B, \tag{A6}$$

the elementary bilinears contribute to the two-band kernel as follows. Direct Fourier transformation gives terms such as $c_A(\mathbf{k})c_B(-\mathbf{k})$; Table II also gives the relabeled form used to read the kernel, with $c_X(\mathbf{k})^\dagger = c_X(-\mathbf{k})$.

TABLE II. Fourier rules for the mean-field kernel. The off-diagonal term is read after relabeling momenta into $ic_A(-\mathbf{k})c_B(\mathbf{k})$.

Term	Fourier form	Kernel
$\lambda\, ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\delta})$	$i\lambda e^{-i\mathbf{k}\cdot\boldsymbol{\delta}} c_A(\mathbf{k})c_B(-\mathbf{k})$ $= i\lambda e^{+i\mathbf{k}\cdot\boldsymbol{\delta}} c_A(-\mathbf{k})c_B(\mathbf{k})$	$\Delta\Phi = \lambda e^{+i\mathbf{k}\cdot\boldsymbol{\delta}}$
$\lambda\, ic_B(\mathbf{r})c_A(\mathbf{r}+\boldsymbol{\delta})$	$-i\lambda e^{+i\mathbf{k}\cdot\boldsymbol{\delta}} c_A(\mathbf{k})c_B(-\mathbf{k})$ $= -i\lambda e^{-i\mathbf{k}\cdot\boldsymbol{\delta}} c_A(-\mathbf{k})c_B(\mathbf{k})$	$\Delta\Phi = -\lambda e^{-i\mathbf{k}\cdot\boldsymbol{\delta}}$
$\lambda\, ic_X(\mathbf{r})c_X(\mathbf{r}+\boldsymbol{\delta})$	$-2\lambda\sin(\mathbf{k}\cdot\boldsymbol{\delta})c_X(-\mathbf{k})c_X(\mathbf{k})$	$\Delta\mathcal{H}_{XX} = -2\lambda\sin(\mathbf{k}\cdot\boldsymbol{\delta})$

The direct channel gives two identical A–B hoppings on the original α bond,

$$H_P = 2J\sum_{\alpha,\mathbf{r}} P_\alpha\, ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\Delta}_\alpha) - NJ\sum_\alpha P_\alpha^2. \tag{A7}$$

Using Table II,

$$\Phi_P(\mathbf{k}) = (K + 2JP_x) + (K + 2JP_y)e^{-i\mathbf{k}\cdot\mathbf{R}_1} + (K + 2JP_z)e^{-i\mathbf{k}\cdot\mathbf{R}_2}. \tag{A8}$$

The exchange channel gives oriented same-sublattice hoppings along $\mathbf{p}_\alpha$. With the staggered convention in Eq. (A5), Table II gives the diagonal terms

$$\mathcal{H}_Q(\mathbf{k}) = \begin{pmatrix} -m_J(\mathbf{k}) & 0 \\ 0 & m_J(\mathbf{k}) \end{pmatrix}, \tag{A9}$$

where

$$m_J(\mathbf{k}) = 2J\left[Q_z \sin(\mathbf{k}\cdot\mathbf{R}_1) + Q_x \sin\{\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\} + Q_y \sin(\mathbf{k}\cdot\mathbf{R}_2)\right]. \tag{A10}$$

The diagonal mass including the diagnostic chiral hopping is

$$\begin{aligned} m(\mathbf{k}) = m_J(\mathbf{k}) + m_\kappa(\mathbf{k}) =& 2J\left[Q_z \sin(\mathbf{k}\cdot\mathbf{R}_1) + Q_x \sin\{\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\} + Q_y \sin(\mathbf{k}\cdot\mathbf{R}_2)\right] \\ &+ 2\kappa\left[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\{\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\} - \sin(\mathbf{k}\cdot\mathbf{R}_2)\right]. \end{aligned} \tag{A11}$$

For the crossed inter-sublattice channel, the residual A–B and B–A hoppings are summarized in Table III. Collecting

TABLE III. Cross-channel terms in the off-diagonal kernel. The $B \to A$ terms include the Majorana-swap sign in Table II.

Pair	$B \to A$	$A \to B$
x	$-JS_1 e^{-i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)}$	$-JS_2 e^{+i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)}$
y	$-JS_2 e^{-i\mathbf{k}\cdot(\mathbf{R}_1 + \mathbf{R}_2)}$	$-JS_3 e^{-i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)}$
z	$-JS_1 e^{-i\mathbf{k}\cdot(\mathbf{R}_1 + \mathbf{R}_2)}$	$-JS_3 e^{+i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)}$

equal phase factors gives

$$\begin{aligned} \Phi_S(\mathbf{k}) =& -J(S_2 + S_3)e^{+i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)} - J(S_1 + S_3)e^{-i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)} \\ &- J(S_1 + S_2)e^{-i\mathbf{k}\cdot(\mathbf{R}_1 + \mathbf{R}_2)}. \end{aligned} \tag{A12}$$

Combining the direct and crossed inter-sublattice channels,

$$\begin{aligned} \Phi(\mathbf{k}) =& (K + 2JP_x) + (K + 2JP_y)e^{-i\mathbf{k}\cdot\mathbf{R}_1} + (K + 2JP_z)e^{-i\mathbf{k}\cdot\mathbf{R}_2} \\ &- J(S_2 + S_3)e^{+i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)} - J(S_1 + S_3)e^{-i\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)} \\ &- J(S_1 + S_2)e^{-i\mathbf{k}\cdot(\mathbf{R}_1 + \mathbf{R}_2)}. \end{aligned} \tag{A13}$$

The quadratic Hamiltonian is therefore

$$\mathcal{H}(\mathbf{k}) = \begin{pmatrix} -m(\mathbf{k}) & i\Phi(\mathbf{k}) \\ -i\Phi^*(\mathbf{k}) & m(\mathbf{k}) \end{pmatrix}, \qquad \omega(\mathbf{k}) = \sqrt{m(\mathbf{k})^2 + |\Phi(\mathbf{k})|^2}. \tag{A14}$$

At the $\kappa = 0$ saddle point, the energy density is

$$\varepsilon_{\mathrm{MF}} = -J\sum_\alpha P_\alpha^2 - J\sum_\alpha Q_\alpha^2 + J(S_1S_2 + S_1S_3 + S_2S_3) - \frac{1}{N_k}\sum_{\mathbf{k}} \omega_0(\mathbf{k}), \tag{A15}$$

where $\omega_0(\mathbf{k}) = \sqrt{m(\mathbf{k})|_{\kappa=0}^2 + |\Phi(\mathbf{k})|^2}$.

Appendix B: Mean-Field Parameters Self-Consistent Update

The self-consistency equations are obtained by evaluating the bilinears in Eq. (A5) in the occupied lower band of the $\kappa = 0$ Hamiltonian. Let

$$\mathcal{H}_0(\mathbf{k}) = \begin{pmatrix} -m_0(\mathbf{k}) & i\Phi(\mathbf{k}) \\ -i\Phi^*(\mathbf{k}) & m_0(\mathbf{k}) \end{pmatrix}, \qquad \omega_0(\mathbf{k}) = \sqrt{m_0(\mathbf{k})^2 + |\Phi(\mathbf{k})|^2}. \tag{B1}$$

For the normalized lower-band eigenvector

$$u_-(\mathbf{k}) = \begin{pmatrix} u_A(\mathbf{k}) \\ u_B(\mathbf{k}) \end{pmatrix}, \qquad \mathcal{H}_0(\mathbf{k})u_-(\mathbf{k}) = -\omega_0(\mathbf{k})u_-(\mathbf{k}), \tag{B2}$$

the two component equations give

$$u_A^*(\mathbf{k})u_B(\mathbf{k}) = \frac{i\Phi^*(\mathbf{k})}{2\omega_0(\mathbf{k})}, \qquad |u_A(\mathbf{k})|^2 - |u_B(\mathbf{k})|^2 = \frac{m_0(\mathbf{k})}{\omega_0(\mathbf{k})}. \tag{B3}$$

The output parameters are defined by the same real-space bilinears as the input parameters:

$$\begin{aligned} P_\alpha^{\text{out}} &= \langle ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\Delta}_\alpha)\rangle\,, \\ S_a^{\text{out}} &= \langle ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\delta}_a)\rangle\,, \\ Q_\alpha^{\text{out}} &= \frac{1}{2}\left[\langle ic_A(\mathbf{r})c_A(\mathbf{r}+\mathbf{p}_\alpha)\rangle - \langle ic_B(\mathbf{r})c_B(\mathbf{r}+\mathbf{p}_\alpha)\rangle\right]. \end{aligned} \tag{B4}$$

Here $\boldsymbol{\Delta}_x = \mathbf{0}$, $\boldsymbol{\Delta}_y = -\mathbf{R}_1$, $\boldsymbol{\Delta}_z = -\mathbf{R}_2$; $\boldsymbol{\delta}_1 = \mathbf{R}_1 - \mathbf{R}_2$, $\boldsymbol{\delta}_2 = \mathbf{R}_2 - \mathbf{R}_1$, $\boldsymbol{\delta}_3 = -\mathbf{R}_1 - \mathbf{R}_2$; and $\mathbf{p}_x = \mathbf{R}_1 - \mathbf{R}_2$, $\mathbf{p}_y = \mathbf{R}_2$, $\mathbf{p}_z = \mathbf{R}_1$.

Using the Fourier convention in Eq. (A6), a general inter-sublattice bilinear with cell displacement $\boldsymbol{\delta}$ gives

$$\begin{aligned} \langle ic_A(\mathbf{r})c_B(\mathbf{r}+\boldsymbol{\delta})\rangle &= \frac{i}{N_k}\sum_{\mathbf{k}}\left[e^{i\mathbf{k}\cdot\boldsymbol{\delta}}u_A^*(\mathbf{k})u_B(\mathbf{k}) - e^{-i\mathbf{k}\cdot\boldsymbol{\delta}}u_B^*(\mathbf{k})u_A(\mathbf{k})\right] \\ &= -\frac{2}{N_k}\sum_{\mathbf{k}}\mathrm{Im}\left[e^{i\mathbf{k}\cdot\boldsymbol{\delta}}u_A^*(\mathbf{k})u_B(\mathbf{k})\right] \\ &= -\frac{1}{N_k}\sum_{\mathbf{k}}\frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot\boldsymbol{\delta}}\Phi^*(\mathbf{k})\right]}{\omega_0(\mathbf{k})}. \end{aligned} \tag{B5}$$

Setting $\boldsymbol{\delta} = \boldsymbol{\Delta}_\alpha$ gives the P_α update, while setting $\boldsymbol{\delta} = \boldsymbol{\delta}_a$ gives the S_a update.

For an oriented same-sublattice bilinear,

$$\begin{aligned} \langle ic_X(\mathbf{r})c_X(\mathbf{r}+\mathbf{p}_\alpha)\rangle &= \frac{i}{N_k}\sum_{\mathbf{k}}\left[e^{i\mathbf{k}\cdot\mathbf{p}_\alpha} - e^{-i\mathbf{k}\cdot\mathbf{p}_\alpha}\right]|u_X(\mathbf{k})|^2 \\ &= -\frac{2}{N_k}\sum_{\mathbf{k}}\sin(\mathbf{k}\cdot\mathbf{p}_\alpha)|u_X(\mathbf{k})|^2, \qquad X = A, B. \end{aligned} \tag{B6}$$

Using the sublattice-staggered definition in Eq. (B4),

$$\begin{aligned} Q_\alpha^{\text{out}} &= -\frac{1}{N_k}\sum_{\mathbf{k}}\sin(\mathbf{k}\cdot\mathbf{p}_\alpha)\left[|u_A(\mathbf{k})|^2 - |u_B(\mathbf{k})|^2\right] \\ &= -\frac{1}{N_k}\sum_{\mathbf{k}}\sin(\mathbf{k}\cdot\mathbf{p}_\alpha)\frac{m_0(\mathbf{k})}{\omega_0(\mathbf{k})}. \end{aligned} \tag{B7}$$

Combining Eqs. (B5) and (B7), the fixed-point map is

$$P_\alpha^{\text{out}} = -\frac{1}{N_k}\sum_{\mathbf{k}}\frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot\boldsymbol{\Delta}_\alpha}\Phi^*(\mathbf{k})\right]}{\omega_0(\mathbf{k})}, \qquad \alpha = x, y, z,$$

$$S_a^{\rm out} = -\frac{1}{N_k}\sum_{\mathbf{k}} \frac{\mathrm{Re}\left[e^{i\mathbf{k}\cdot\boldsymbol{\delta}_a}\Phi^*(\mathbf{k})\right]}{\omega_0(\mathbf{k})}, \qquad a=1,2,3,$$
$$Q_\alpha^{\rm out} = -\frac{1}{N_k}\sum_{\mathbf{k}} \sin(\mathbf{k}\cdot\mathbf{p}_\alpha)\frac{m_0(\mathbf{k})}{\omega_0(\mathbf{k})}, \qquad \alpha = x,y,z. \tag{B8}$$

A self-consistent solution satisfies

$$X = X^{\rm out}(X), \qquad X = (P_z, P_x, P_y, Q_z, Q_x, Q_y, S_1, S_2, S_3). \tag{B9}$$

The mass m_κ is added only after this $\kappa = 0$ fixed point is obtained.

Appendix C: Γ-Quadratic Band Touching

We consider the C_3-symmetric, time-reversal-symmetric saddle point at $\kappa = 0$:

$$P_x = P_y = P_z \equiv P, \qquad S_1 = S_2 = S_3 \equiv S, \qquad Q_\alpha = 0.$$

The Hamiltonian is off diagonal,

$$\mathcal{H}_0(\mathbf{k}) = \begin{pmatrix} 0 & i\Phi(\mathbf{k}) \\ -i\Phi^*(\mathbf{k}) & 0 \end{pmatrix}, \qquad E_\pm(\mathbf{k}) = \pm|\Phi(\mathbf{k})|. \tag{C1}$$

Thus the nodes are zeros of $\Phi(\mathbf{k})$.

Let

$$\widetilde{K} = K + 2JP.$$

The C_3-symmetric off-diagonal kernel is

$$\begin{aligned}\Phi(\mathbf{k}) = {}& \widetilde{K}\left[1 + e^{-i\mathbf{k}\cdot\mathbf{R}_1} + e^{-i\mathbf{k}\cdot\mathbf{R}_2}\right] \\ & - 2JS\left[e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}\right].\end{aligned} \tag{C2}$$

Equivalently,

$$\Phi(\mathbf{k}) = \widetilde{K}A(\mathbf{k}) - 2JSB(\mathbf{k}), \tag{C3}$$

where

$$\begin{aligned}A(\mathbf{k}) &= 1 + e^{-i\mathbf{k}\cdot\mathbf{R}_1} + e^{-i\mathbf{k}\cdot\mathbf{R}_2}, \\ B(\mathbf{k}) &= e^{+i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + e^{-i\mathbf{k}\cdot(\mathbf{R}_1-\mathbf{R}_2)} + e^{-i\mathbf{k}\cdot(\mathbf{R}_1+\mathbf{R}_2)}.\end{aligned} \tag{C4}$$

At Γ,

$$\Phi(\Gamma) = 3\delta_\Gamma, \qquad \delta_\Gamma = \widetilde{K} - 2JS = K + 2J(P-S). \tag{C5}$$

Set $\mathbf{k} = \Gamma + \mathbf{q}$, and define

$$a = \mathbf{q}\cdot\mathbf{R}_1, \qquad b = \mathbf{q}\cdot\mathbf{R}_2.$$

The two form factors expand as

$$\begin{aligned}A(a,b) &= 3 - i(a+b) - \frac{1}{2}(a^2+b^2) + \frac{i}{6}(a^3+b^3) + O(q^4), \\ B(a,b) &= 3 - i(a+b) - \left[(a-b)^2 + \frac{1}{2}(a+b)^2\right] + \frac{i}{6}(a+b)^3 + O(q^4).\end{aligned} \tag{C6}$$

Thus

$$A(a,b) - B(a,b) = a^2 - ab + b^2 - \frac{i}{2}ab(a+b) + O(q^4). \tag{C7}$$

Using $\Phi = \delta_\Gamma A + 2JS(A - B)$, the terms controlling the local reconstruction are

$$\begin{aligned}\Phi(\Gamma + \mathbf{q}) =& 3\delta_\Gamma - i\delta_\Gamma(a+b) \\ &+ 2JS(a^2 - ab + b^2) - iJSab(a+b) + \cdots .\end{aligned} \tag{C8}$$

Terms such as $\delta_\Gamma q^2$ are higher order in the splitting analysis because the emerging cones satisfy $q^2 \sim |\delta_\Gamma/(JS)|$.

At $\delta_\Gamma = 0$, the constant and linear terms vanish. Since

$$a = \sqrt{3}q_x, \qquad b = \frac{\sqrt{3}}{2}q_x + \frac{3}{2}q_y,$$

one has

$$a^2 - ab + b^2 = \frac{9}{4}(q_x^2 + q_y^2). \tag{C9}$$

Therefore

$$\Phi(\Gamma + \mathbf{q})\big|_{\delta_\Gamma = 0} = \frac{9JS}{2}(q_x^2 + q_y^2) + O(q^3), \tag{C10}$$

which gives a quadratic band touching at Γ.

For $\delta_\Gamma \neq 0$, write $\Phi = f + ig$. To leading order,

$$\begin{aligned} f(a,b) &= 3\delta_\Gamma + 2JS(a^2 - ab + b^2), \\ g(a,b) &= -(a+b)(\delta_\Gamma + JSab). \end{aligned} \tag{C11}$$

The nodal condition $f = g = 0$ gives

$$3\delta_\Gamma + 2JS(a^2 - ab + b^2) = 0, \tag{C12}$$

together with either

$$a + b = 0 \qquad \text{or} \qquad \delta_\Gamma + JSab = 0. \tag{C13}$$

Real solutions exist for

$$\frac{\delta_\Gamma}{JS} < 0. \tag{C14}$$

In Cartesian momentum coordinates, the six solutions are

$$\begin{aligned} \mathbf{q}_n &= r\left(\cos\frac{n\pi}{3}, \sin\frac{n\pi}{3}\right), \qquad n = 0, \ldots, 5, \\ r &= \left(-\frac{2\delta_\Gamma}{3JS}\right)^{1/2}. \end{aligned} \tag{C15}$$

Thus the Γ-point quadratic touching splits into six simple zeros surrounding Γ.

Appendix D: Chern Number

We now add the small diagnostic mass m_κ to the six cones obtained in Appendix C. The calculation is local in momentum space and determines the change in the lower-band Chern number across the Γ-point reconstruction.

The winding of each cone is fixed by the Jacobian of the map $(a,b) \mapsto (f,g)$. From Eq. (C11),

$$\partial_a f = 2JS(2a - b), \qquad \partial_b f = 2JS(-a + 2b),$$

$$\partial_a g = -(\delta_\Gamma + JSab) - JSb(a+b),$$
$$\partial_b g = -(\delta_\Gamma + JSab) - JSa(a+b). \tag{D1}$$

At the six zeros $\mathbf{q}_n$, the determinant is

$$\mathcal{J}_n = \det \frac{\partial(f,g)}{\partial(a,b)}\bigg|_{\mathbf{q}=\mathbf{q}_n} = -(-1)^n \frac{27\sqrt{3}}{2}(JS)^2 r^3. \tag{D2}$$

The transformation from (q_x, q_y) to (a, b) has positive determinant,

$$\det \frac{\partial(a,b)}{\partial(q_x,q_y)} = \frac{3\sqrt{3}}{2} > 0,$$

so the orientation is unchanged. With the winding convention used in the main text,

$$w_n = (-1)^n. \tag{D3}$$

The diagnostic mass is

$$m_\kappa(\mathbf{k}) = 2\kappa\big[\sin(\mathbf{k}\cdot\mathbf{R}_1) - \sin\{\mathbf{k}\cdot(\mathbf{R}_1 - \mathbf{R}_2)\} - \sin(\mathbf{k}\cdot\mathbf{R}_2)\big]. \tag{D4}$$

Near Γ,

$$\begin{aligned} m_\kappa(a,b) &= 2\kappa\,[\sin a - \sin(a-b) - \sin b] \\ &= -\kappa\, ab(a-b) + O(q^5) \\ &= -\frac{3\sqrt{3}}{4}\kappa\, q_x(q_x^2 - 3q_y^2) + O(q^5). \end{aligned} \tag{D5}$$

For $\mathbf{q}_n = r(\cos n\pi/3, \sin n\pi/3)$,

$$\operatorname{sgn} m_n = \operatorname{sgn} m_\kappa(\mathbf{q}_n) = -\operatorname{sgn}(\kappa)(-1)^n. \tag{D6}$$

A massive simple zero contributes to the lower-band Chern number as

$$C_n = -\frac{1}{2} w_n \operatorname{sgn} m_n. \tag{D7}$$

Using Eqs. (D3) and (D6),

$$\begin{aligned} C_n &= -\frac{1}{2}(-1)^n\,[-\operatorname{sgn}(\kappa)(-1)^n] \\ &= \frac{1}{2}\operatorname{sgn}(\kappa). \end{aligned} \tag{D8}$$

Thus, for $\kappa > 0$,

$$\Delta\mathcal{C}_\kappa = \sum_{n=0}^{5} C_n = 3. \tag{D9}$$

The full Brillouin-zone calculation fixes the absolute values quoted in the main text. The $\kappa = 0$ self-consistent spectra remain gapless; $\mathcal{C}_\kappa$ refers to the Hamiltonian after the small m_κ term is added.

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